

ECB1332 - DESIGN PROJECT
EVENT-BASED SMART WATER PIPELINE MONITORING
AND AUTOMATIC VALVE CONTROL SYSTEM

A PROJECT REPORT

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BONAFIDE CERTIFICATE

Certified that this project report “**EVENT-BASED SMART WATER PIPELINE MONITORING AND AUTOMATIC VALVE CONTROL SYSTEM**” is the Bonafide work of **ABITHA BANU M(927623BEC003), BHARATHI S (927623BEC018), BRINDHASRI I(927623BEC019), DEEPIKA S(927623BEC023), DEVI S T(927623BEC025), DHANYA D(927623BEC027)** who carried out the project work during the academic year 2025- 2026 under my supervision.

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ABSTRACT

Water distribution systems play a vital role in supplying water to residential, commercial, and institutional infrastructures. However, leakage and unauthorized water usage in pipeline networks lead to significant water loss, economic damage, and inefficient resource management. To address these challenges, this project proposes an Event-Based Smart Water Pipeline Leakage Detection System using Internet of Things (IoT) technology. The system continuously monitors water flow and pressure parameters using flow sensors and pressure sensors installed at different levels of the pipeline network, including the main pipeline and sub-pipelines. A microcontroller-based unit processes real-time sensor data to detect abnormalities by comparing inlet and outlet flow rates along with pressure variations. Based on these parameters, the system intelligently classifies events into normal conditions, leakage, or water theft. Upon detection of any abnormal event, the system generates location-specific alerts. In the case of leakage in the main pipeline, notifications are sent to the municipal authority, whereas leakage in sub-pipelines triggers alerts to the respective water management department. Additionally, the system incorporates an automated control mechanism using solenoid valves, which are activated to shut off the water supply during leakage, thereby preventing further loss and damage. This event-driven approach ensures efficient water management, reduces manual monitoring, and enhances response time through real-time alerts and automatic control.

Abstract (Key words)	POs Mapping
Smart Water Monitoring, Leakage Detection, Flow Sensor, Pressure Sensor, ESP32, IoT, Pipeline Monitoring, Water Theft Detection, Automation, Real-time Alert System	PO1, PO2, PO3, PO4, PO5, PO6, PO7, PO8, PO9, PO10, PO11, PO12, PSO1, PSO2

SDG Goal		Remarks
SDG 6 	This project aligns with focusing on conserving water and ensuring efficient and sustainable water management.	Enhances detecting water leakage and preventing water loss for smart and sustainable Water management.

Project Component	Relevant IEEE Standards
Flow Sensors (Inlet & Outlet)	IEEE 1451
Pressure Sensors	IEEE 1457
Fiber Optic Cable	IEEE 802.3
ESP32 / Arduino Microcontroller	IEEE 802.15.4
Solenoid Valve	IEEE C37
Self-Healing Coating Capsules	IEEE 1012
Documentation & testing	IEEE 830, IEEE 829

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LIST OF ABBREVAITIONS

ACRONYM		ABBREVIATION
AC	-	Alternating Current
ADC	-	Analog to Digital Converter
API	-	Application Programming Interface
ESP32	-	Espressif 32-bit Microcontroller
GPS	-	Global Positioning System
HTTP	-	Hypertext Transfer Protocol
MCU	-	Microcontroller Unit
MQTT	-	Message Queuing Telemetry Transport
PCB	-	Printed Circuit Board
SMS	-	Short Message Service
UI	-	User Interface
Wi-Fi	-	Wireless Fidelity

CHAPTER 1

INTRODUCTION

Water is one of the most essential natural resources for human life, agriculture, and industrial applications. Efficient water distribution and monitoring have become increasingly important due to population growth, urbanization, and water scarcity issues. Traditional water distribution systems rely mainly on manual monitoring and maintenance, which often leads to water leakage, theft, and inefficient usage. These problems result in significant water loss, increased operational costs, and reduced system reliability.

Therefore, there is a need for a smart and automated water monitoring system to ensure efficient water management. Recent advancements in Internet of Things (IoT) and embedded systems have made it possible to monitor and control water distribution systems in real time. IoT-based systems use sensors, microcontrollers, and communication modules to collect data and transmit it to users or authorities. These systems help in early detection of faults, reduction of water wastage, and improvement of system efficiency.

By integrating automation and intelligent monitoring, water distribution can be made more reliable and cost-effective. This project focuses on designing an IoT-based water monitoring system capable of detecting water leakage and theft conditions. The system uses sensors to monitor water flow and automatically identify abnormal conditions. The collected data is transmitted to users for monitoring and control. The proposed system aims to reduce water wastage, improve efficiency, and provide a cost-effective solution for smart water management.

1.1 OBJECTIVE

The primary objective of this project is to design and develop a self-healing coating system using microcapsule technology to automatically repair cracks and damages on material surfaces. The system aims to improve the durability and lifespan of coated components by preventing corrosion, moisture penetration, and environmental degradation. Another objective is to reduce maintenance costs and minimize manual inspection by enabling autonomous healing without human intervention.

The project also focuses on enhancing the reliability and performance of materials used in harsh environments such as industrial equipment, pipelines, and electronic devices. Additionally, the proposed system aims to improve surface protection efficiency and maintain structural integrity over long periods. The use of self-healing capsules helps reduce material failure and increases operational safety.

Furthermore, the objective includes developing a cost-effective and eco-friendly coating solution suitable for various applications such as aerospace, automotive, marine, and infrastructure systems. The system also aims to support sustainable engineering practices by reducing material waste and extending product life. Overall, the project focuses on creating a smart, durable, and efficient self-healing coating system for advanced protection applications.

1.2 PROBLEM STATEMENT

Surface coatings used in industrial equipment, pipelines, and electronic components are prone to damage due to environmental factors such as corrosion, moisture, temperature variation, and mechanical stress. These damages often appear as small cracks or scratches that gradually increase and lead to serious structural failures.

In many cases, minor damages are not detected at an early stage, allowing corrosion and degradation to spread further. This reduces the lifespan of materials and affects system reliability. Additionally, maintenance and repair work in remote or hazardous environments becomes difficult and unsafe. Existing coating solutions mainly focus on protection but do not provide automatic repair functionality. Therefore, there is a need for an intelligent self-healing coating system that can automatically repair minor damages, improve durability, reduce maintenance costs, and enhance system reliability. The proposed self-healing coating capsule system addresses these issues by providing autonomous crack healing and long-term surface protection.

1.2.1 PROJECT DESCRIPTION

The Self-Healing Coating Capsules project focuses on developing an advanced coating system capable of repairing surface damage automatically. The system uses microcapsules embedded within the coating material. These capsules contain healing agents that are released when cracks or scratches occur on the surface.

When damage appears, the microcapsules break and release the healing material into the cracked region. The healing agent reacts with the surrounding environment and fills the damaged area. This process restores the protective layer and prevents further corrosion or degradation. The self-healing process improves durability and reduces maintenance requirements.

The proposed system can be applied in various industries such as pipelines, aerospace, automotive, marine structures, and electronic devices. The system enhances surface protection, increases lifespan, and reduces repair costs. Additionally, the project supports sustainable engineering by minimizing material waste and improving system reliability.

1.2.2 METHODOLOGY OVERVIEW

1. Material Selection:

Select suitable coating materials, healing agents, and encapsulation materials to develop an effective self-healing coating system with improved durability and corrosion resistance.

2. Microcapsule Fabrication:

Prepare self-healing microcapsules using appropriate encapsulation techniques to store healing agents that can repair surface damage automatically.

3. Coating Preparation:

Mix the fabricated microcapsules with the base coating material to form a smart self-healing coating solution.

4. Surface Application:

Apply the prepared self-healing coating onto the target surfaces such as metal structures, pipelines, and industrial components.

5. Damage Formation and Healing:

When cracks or scratches occur due to environmental or mechanical stress, the microcapsules break and release healing agents to repair the damaged area.

6. Performance Analysis:

Evaluate the healing efficiency, corrosion resistance, and durability of the self-healing coating under different operating conditions.

7. Testing and Validation:

Conduct experimental testing to verify the effectiveness, reliability, and long-term performance of the self-healing coating system.

CHAPTER 2

LITERATURE REVIEW

2.1 IOT-BASED SMART WATER LEAKAGE DETECTION SYSTEM USING ARDUINO UNO

Authors: S. Fulawade et al. (2023)

S. Fulawade et al. proposed an IoT-based smart water leakage detection system using Arduino UNO, flow sensors, and cloud connectivity. The system monitors water flow continuously and sends real-time alerts when leakage is detected. The collected data is transmitted to cloud platforms, allowing users to monitor the pipeline remotely. This system improves water management efficiency and reduces manual monitoring. However, the system lacks advanced leak localization techniques and automated corrective actions.

2.2 AI-BASED WATER PIPELINE LEAKAGE DETECTION SYSTEM

Authors: J. Smith et al. (2024)

J. Smith et al. developed an AI-based water pipeline leakage detection system that uses machine learning algorithms to analyze sensor data. The system identifies leakage patterns and classifies the severity of leakage conditions. This improves detection accuracy and enables intelligent decision-making. However, the system depends heavily on training datasets and does not include physical repair or automatic control mechanisms.

2.3 FIBER OPTIC DISTRIBUTED SENSING SYSTEM FOR PIPELINE MONITORING

Authors: L. Zhang (2025)

L. Zhang proposed a fiber optic distributed sensing system for pipeline monitoring. The system uses backscattered light signals to detect vibration and temperature changes in pipelines. This approach provides high accuracy in leak detection and helps identify the exact location of pipeline faults. However, the system is costly and lacks IoT-based automated control features.

2.4 IOT-BASED PIPELINE LEAKAGE DETECTION USING PRESSURE SENSORS

Authors: S. K. Patel (2025)

S. K. Patel introduced an IoT-based pipeline leakage detection system using pressure sensors. The system detects leakage by monitoring abnormal pressure drops in the pipeline. When pressure variations occur, alerts are sent to users for immediate action. This improves detection efficiency and reduces water loss. However, the system does not support automatic valve control or self-healing capabilities.

2.5 OPTICAL FIBER SPECKLE PATTERN LEAKAGE DETECTION SYSTEM

Authors: Y. Chen (2024)

Y. Chen presented an optical fiber speckle pattern leakage detection system for detecting small pipeline leaks. The system provides high sensitivity and continuous monitoring of water pipelines. This method improves detection accuracy, especially for minor leakage conditions.

2.6 SMART WATER DISTRIBUTION SYSTEM USING IOT AND CLOUD COMPUTING

Authors: R. Gupta (2025)

R. Gupta proposed a smart water distribution system using IoT and cloud computing technology. The system monitors water flow and detects leakage conditions in real time. Cloud connectivity enables remote monitoring and improves water distribution efficiency. However, the system lacks real-time physical intervention mechanisms such as automatic valve control.

SUMMARY

The literature survey highlights various water leakage detection systems using IoT, AI, and fiber optic technologies. These systems improve monitoring accuracy and reduce water wastage. However, many systems lack automation, intelligent classification, and cost-effective implementation. Therefore, there is a need for a smart, automated, and efficient water monitoring system that can detect leakage and water theft with real-time control mechanisms

CHAPTER 3

EXISTING SYSTEM

The existing water monitoring system mainly depends on traditional methods and basic sensor-based monitoring. In many water distribution systems, manual inspection is used to identify leakage and unauthorized water usage. These methods are time-consuming and inefficient, especially in large-scale pipeline networks. Due to the lack of real-time monitoring, faults are often detected only after significant water loss has occurred. This results in wastage of water, increased maintenance costs, and poor system performance. Some existing systems use IoT technology and sensors to monitor water flow and pressure conditions. These systems provide remote monitoring and basic alert mechanisms. However, many of them focus only on monitoring and do not include automatic control or proper fault classification. As a result, the efficiency of water management remains limited.

3.1 MANUAL MONITORING SYSTEM

The Traditional water monitoring systems rely heavily on manual inspection to detect leakage and faults in water distribution pipelines. Field workers are required to physically inspect pipelines and distribution points at regular intervals. This process is time-consuming and inefficient, especially for large-scale water networks. In many cases, small leaks remain unnoticed until they become major issues, leading to significant water loss. Additionally, manual inspection increases operational cost, requires more manpower, and reduces the overall efficiency of water management systems

3.2 BASIC SENSOR -BASED MONITORING

Some existing systems use basic sensors such as flow sensors and pressure sensors to monitor water distribution. These sensors help detect abnormal variations in water flow and pressure levels. However, most of these systems only provide monitoring capabilities without automated control mechanisms. When faults occur, manual intervention is still required to take corrective action. Furthermore, these systems often lack advanced data analysis and fault classification, which reduces their effectiveness in identifying the exact cause of water loss.

3.3 LIMITED IOT-BASED MONITORING

Certain modern systems incorporate IoT technology to enable remote monitoring of water distribution networks. These systems allow users to view water flow and pressure data through cloud platforms or mobile applications. Although IoT-based systems provide alerts when abnormal conditions occur, they often lack intelligent decision-making capabilities. Many systems only notify users without automatically controlling water flow or preventing leakage. This limitation reduces the overall performance and efficiency of water monitoring systems.

3.4 LACK OF REAL-TIME MONITORING

Most existing water monitoring systems do not provide continuous real-time monitoring. Data is often collected at fixed intervals or through manual inspection, which delays fault detection. Due to this delay, leakage and unauthorized water usage may continue for long periods before being identified. This results in excessive water wastage, increased maintenance costs, and poor system performance.

3.5 LACK OF FAULT CLASSIFICATION SYSTEM

Many existing water monitoring systems are unable to differentiate between leakage and unauthorized water usage. When abnormal flow or pressure variation occurs, the system only indicates a fault without identifying the exact cause. This makes it difficult for maintenance teams to take appropriate corrective actions. In some cases, leakage may be mistaken for water theft or vice versa, leading to inefficient fault handling. The absence of proper fault classification reduces system reliability and increases maintenance time. Therefore, an improved system is required to accurately classify leakage and water theft conditions for better water management.

3.5.1 BLOCK DIAGRAM REPRESENTATION

This diagram shows a basic monitoring system that starts with sensors collecting data from equipment or processes. The collected data is sent to a manual monitoring unit where staff review the information regularly. If any abnormal condition is detected, the system sends alerts through a simple alert mechanism. In addition, manual inspection is carried out to physically check conditions and support the monitoring process.

The Fig.3.1 shows the representation of block diagram of Existing system

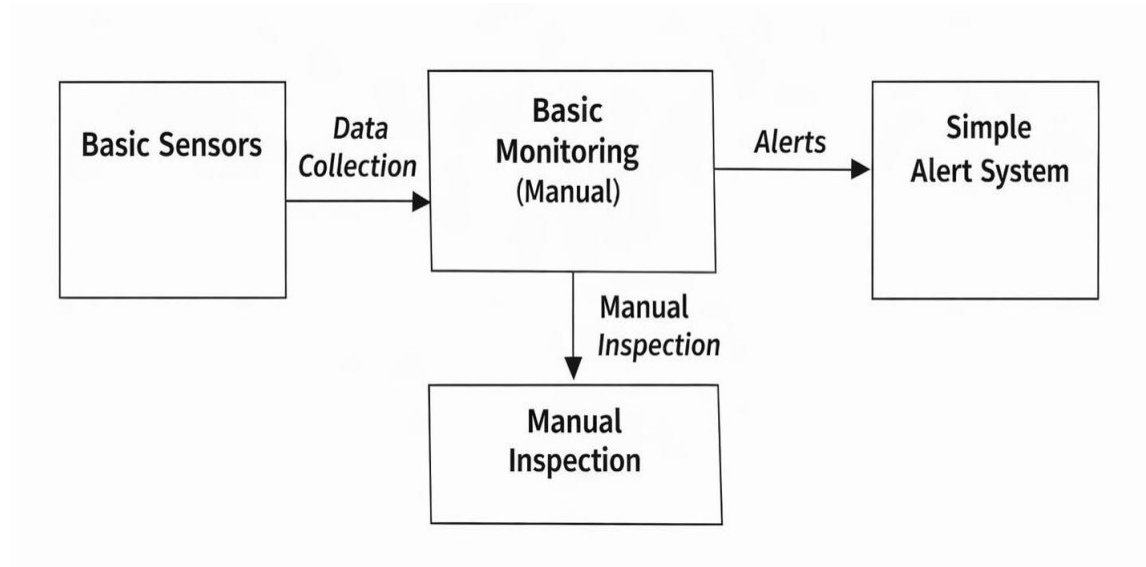


Fig:3.1 BLOCK DIAGRAM OF EXISTING SYSTEM

3.6 LIMITATIONS OF EXISTING SYSTEM

1. Most existing water monitoring systems do not provide real-time fault detection, causing delays in identifying leakage and water theft.
2. Manual monitoring requires more human effort, increases operational costs, and lowers overall efficiency.
3. Many systems cannot clearly differentiate between leakage and theft, making it difficult to find the real cause of water loss.
4. Some systems need complex hardware setups and advanced network infrastructure, increasing installation and maintenance expenses.
5. High power consumption in certain systems reduces reliability and affects long-term continuous operation.
6. Lack of automated control mechanisms prevents immediate action during abnormal conditions, leading to more water wastage.

CHAPTER 4

PROPOSED SYSTEM

4.1 INTRODUCTION

The proposed system is designed to detect water leakage and water theft in pipelines using IoT and embedded technology. It uses two flow sensors placed at the inlet and outlet of the pipeline to continuously measure water flow. The collected data is sent to a microcontroller, which compares the input and output flow values to identify any discrepancies. Based on this comparison, the system classifies the condition as normal operation, leakage, or water theft. To improve accuracy and efficiency, the system can also integrate additional sensors such as pressure sensors to detect variations within the pipeline.

All sensor data is processed by the microcontroller and transmitted to a cloud platform using communication modules like Wi-Fi or GSM. This enables real-time monitoring of water usage through a mobile or web application, along with access to historical data and usage analysis. When abnormal conditions are detected, the system automatically sends alerts to users via notifications or messages and takes immediate action by controlling the valve to stop water flow.

This helps prevent water loss and reduces damage. The system is cost-effective, energy-efficient, and scalable, making it suitable for residential, industrial, and municipal water distribution systems while promoting smart water management. Furthermore, the system can be enhanced with advanced data analytics and machine learning techniques to predict potential leakage points and detect unusual usage patterns over time. This predictive capability allows for preventive maintenance, reducing operational costs and improving system reliability.

4.2 BLOCK DIAGRAM OF PROPOSED SYSTEM

The Fig.4.1 shows the representation of block diagram of proposed system

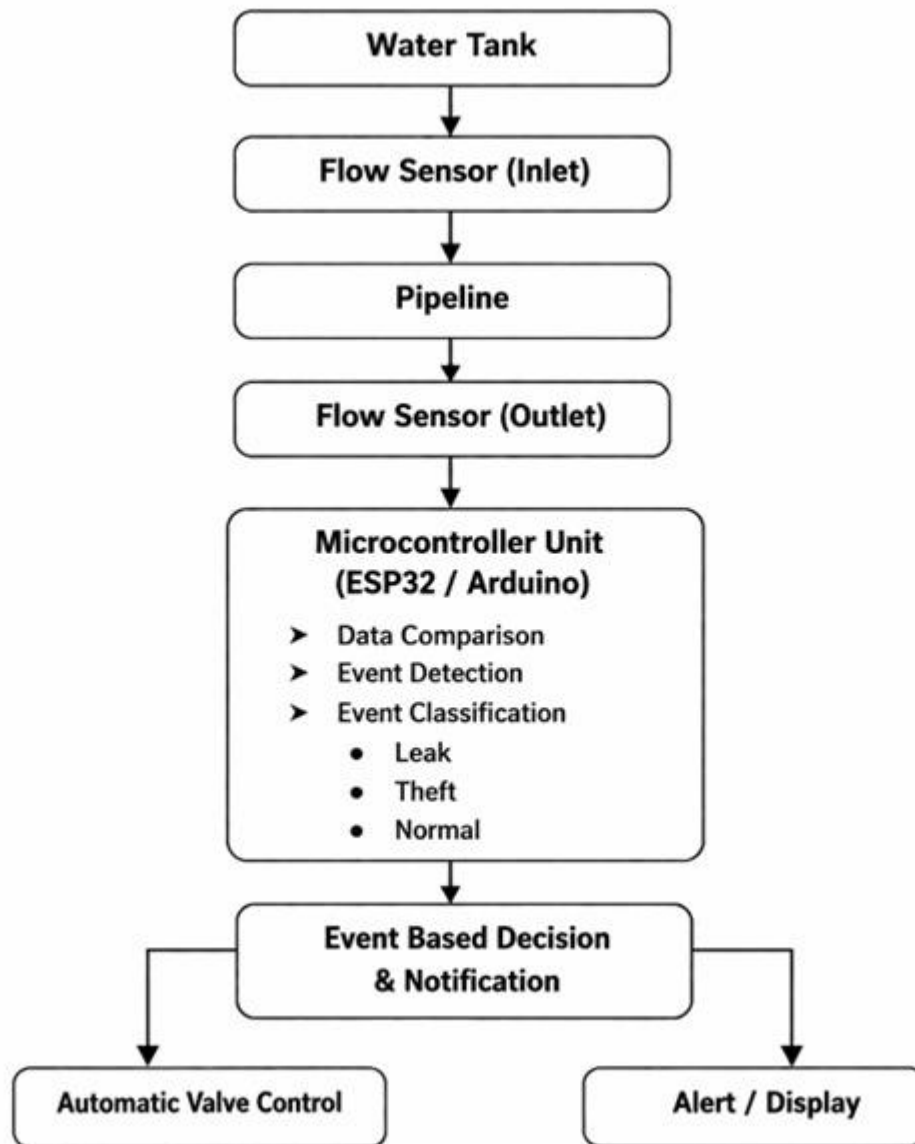


FIG.4.1. BLOCK DIAGRAM OF PROPOSED SYSTEM

4.2.1 WORKING PRINCIPLE

Step 1: Water from the tank enters the pipeline, and the inlet flow sensor measures the incoming water flow

Step 2: Water passes through the pipeline, and the outlet flow sensor measures the outgoing water flow.

Step 3: Both sensor readings are sent to the microcontroller unit (ESP32/Arduino).

Step 4: The microcontroller compares the inlet and outlet flow data to detect any abnormal difference.

Step 5: Based on the comparison, the system identifies the condition as Normal, Leakage, or Theft.

Step 6: If an abnormal condition is detected, the controller makes an event-based decision automatically.

Step 7: The automatic valve control operates to stop or reduce water flow if needed.

Step 8: Alerts or notifications are displayed to inform the user for immediate action.

4.3 COMPONENTS USED

1.FLOW SENSOR

The flow sensor which is shown in Fig. 4.3.1 measures the water flow rate in the pipeline. It helps in monitoring the amount of water entering and leaving the system. The collected data is sent to the microcontroller for analysis and comparison to detect leakage or unauthorized water usage.



Fig.4.3.1 FLOW SENSOR

2. PRESSURE SENSOR

The pressure sensor which is shown in Fig.4.3.2 monitors the pressure level inside the pipeline. Sudden pressure drops or abnormal variations indicate possible leakage or pipe damage. The sensor continuously sends pressure data to the microcontroller for fault detection.



Fig.4.3.2 PRESSURE SENSOR

3. ESP32 MICROCONTROLLER

The ESP32 microcontroller which is shown in Fig.4.3.3 acts as the main control unit of the system. It receives inputs from flow sensors and pressure sensors, processes the data, and detects leakage or water theft. It also controls automatic valve operation and sends alerts to users through IoT communication.

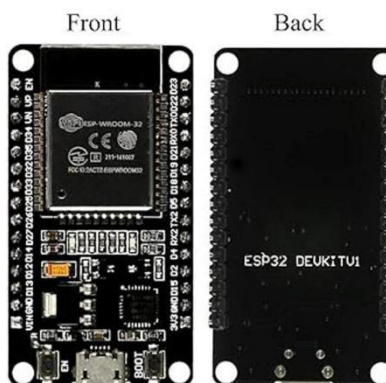


Fig.4.3.3 ESP32 Microcontroller

4. WATER PIPELINE

The water pipeline which is shown in Fig.4.3.4 acts as the medium for water transportation. Sensors are placed along the pipeline to monitor flow rate and pressure conditions. Any abnormal variation in water flow inside the pipeline helps in identifying leakage or unauthorized water usage.



Fig.4.3.4 WATER PIPELINE

5. IOT COMMUNICATION MODULE

The IoT communication module which is shown in Fig.4.2.5 enables wireless communication between the system and the user. It sends real-time sensor data, leakage alerts, and system status to cloud platforms or mobile applications for remote monitoring.



Fig.4.3.5 IOT COMMUNICATION MODULE

6. CLOUD DASHBOARD

The cloud dashboard which is shown in Fig.4.3.6 stores and displays real-time monitoring data. Users can access the system status, water usage, leakage alerts, and historical data through the cloud dashboard from anywhere.



Fig.4.3.6 CLOUD DASHBOARD

7. ALERT NOTIFICATION SYSTEM

The alert notification system which is shown in Fig.4.2.7 sends warning messages to users when leakage or water theft is detected. Alerts can be sent through mobile notification, SMS, or email to ensure immediate action.

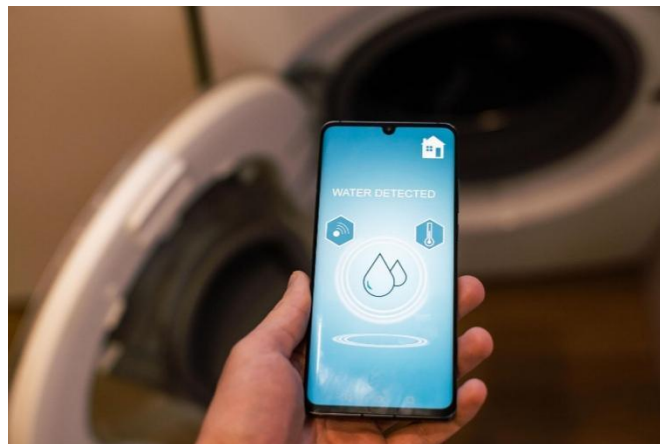


Fig.4.3.7 ALERT NOTIFICATION SYSTEM

8. WATER TANK SYSTEM

The water tank which is shown in Fig.4.3.8 acts as the primary water storage unit. The system monitors water flow from the tank to the pipeline using sensors. This helps in identifying abnormal water consumption and leakage.



Fig.4.3.8 WATER TANK SYSTEM

9. COMPLETE HARDWARE SETUP

The complete hardware setup which is shown in Fig.4.2.9 consists of flow sensors, pressure sensors, ESP32 microcontroller, automatic valve control, IoT module, and water pipeline. All components work together to monitor water flow, detect leakage or theft, and send real-time alerts.

CHAPTER 5

RESULTS AND DISCUSSION

The cloud dashboard displays real-time monitoring data collected from the inlet and outlet flow sensors. The sensor values are continuously transmitted to the cloud platform using IoT connectivity. The dashboard visualizes water flow rates, system status, and alert notifications for leakage and theft detection. During normal operation, the inlet and outlet flow values remain nearly equal, and the dashboard indicates a normal system condition. When leakage occurs, the inlet flow value becomes higher than the outlet value. The cloud dashboard immediately displays abnormal conditions and generates alert notifications. Similarly, during water theft conditions, unusual flow patterns and irregular consumption values are observed.

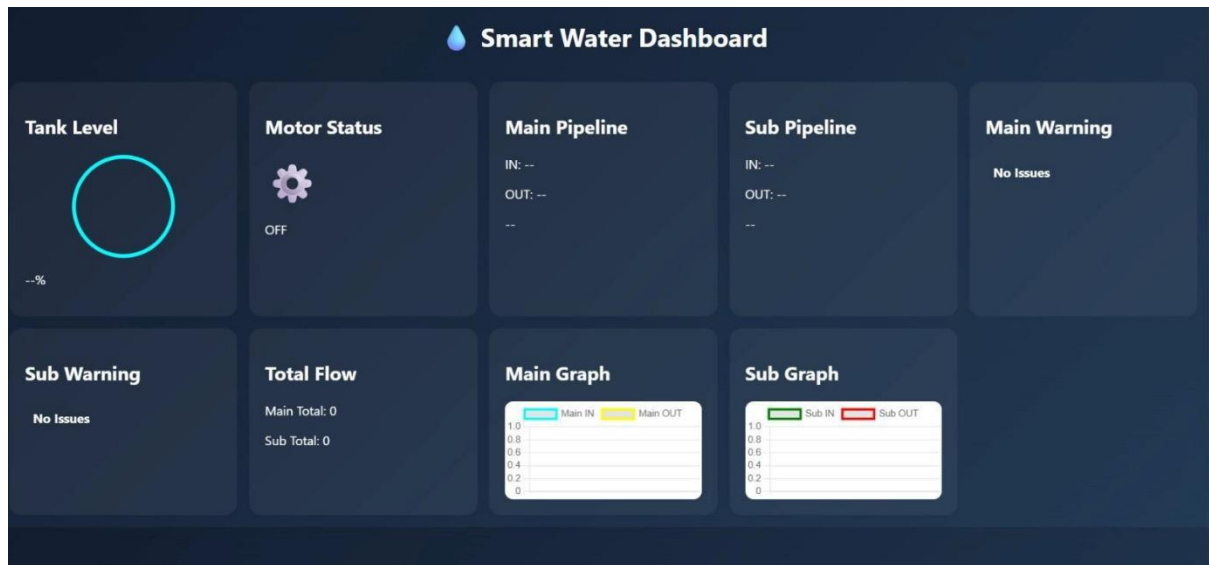


Fig.5.1. DASHBOARD FOR WATER MONITORING SYSTEM

The notification system provides real-time alerts to users when abnormal conditions are detected in the pipeline. When leakage or water theft occurs, the microcontroller

sends data to the cloud platform, which immediately generates alert notifications. These notifications are sent to mobile devices or monitoring systems for quick response. During normal operation, no alerts are generated, and the system continues monitoring water flow. When leakage occurs, the system detects the difference between inlet and outlet flow values and sends a leakage alert notification. This helps users identify the issue quickly and take necessary action. Similarly, when water theft is detected, the system sends a theft alert notification indicating abnormal water usage. These notifications reduce response time and help prevent further water loss

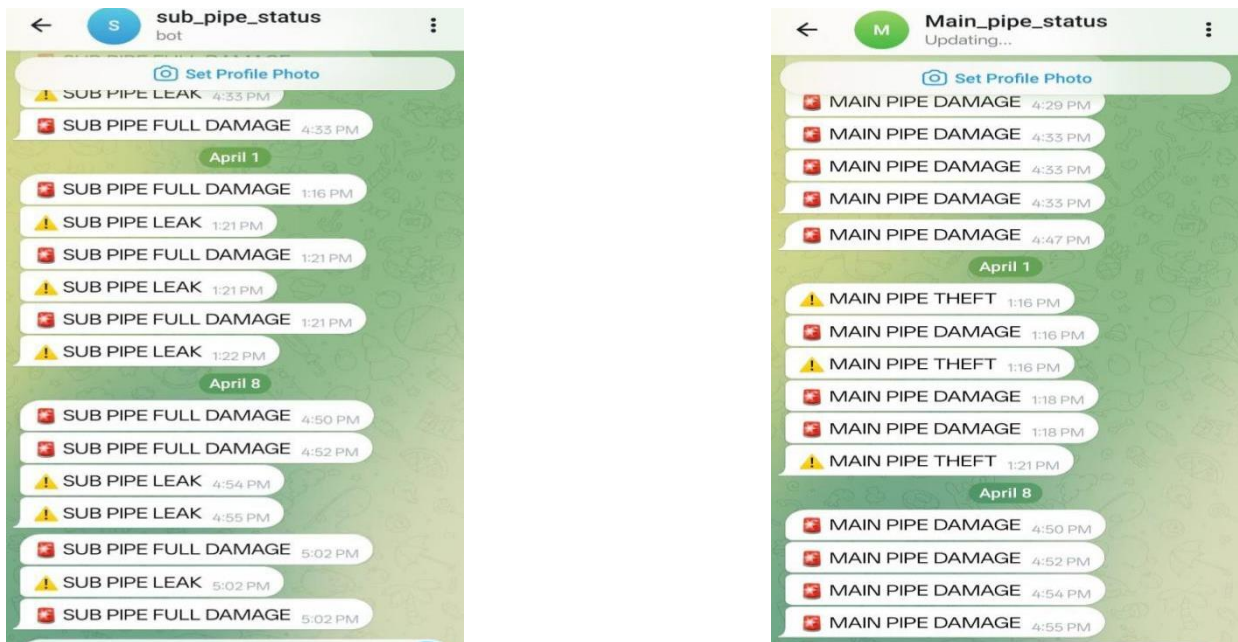


Fig.5.2. NOTIFICATION ALERT RESULTS

CHAPTER 6

CONCLUSION AND FUTURE WORK

CONCLUSION

The proposed smart water monitoring system is successfully developed to detect water leakage and water theft using flow sensors and IoT technology. It continuously monitors inlet and outlet water flow to identify abnormal changes in real time. This comparison method helps in accurate detection of leakage and unauthorized water usage. When an abnormal condition is detected, the system immediately sends alerts to users. At the same time, it automatically controls the valve to stop water flow and reduce water loss. The cloud-based dashboard allows real-time monitoring, data visualization, and remote access from anywhere. The system improves water management by reducing wastage and increasing operational efficiency. It also overcomes the limitations of traditional manual monitoring methods, which are slow and less reliable. The system is cost-effective, energy-efficient, reliable, and suitable for homes, industries, and municipal water networks. Overall, this project provides a practical solution for sustainable water management and supports smart city development.

FUTURE WORK

1. **AI-Based Leakage Prediction:** Artificial Intelligence and Machine Learning can be used to predict leakage before it occurs for early maintenance.
2. **Mobile Application Development:** A mobile app can be developed for real-time monitoring, alerts, and remote valve control.
3. **GPS Location Tracking:** GPS modules can help identify the exact location of leakage or theft in large water networks.

4. **Solar Power Integration:** Solar panels can be used to reduce electricity consumption and improve energy efficiency.
5. **Multiple Sensor Integration:** Additional sensors like water level, humidity, and vibration sensors can improve detection accuracy.
6. **Smart Automatic Valve System:** Advanced valves can isolate only the affected leakage area instead of stopping the full water supply.
7. **Cloud Data Analytics:** Cloud platforms can analyze long-term water usage data and optimize distribution systems.
8. **Industrial Level Implementation:** The system can be expanded for smart cities, industries, and municipal water management.

APPENDICES

ESP32 CODE:

```
#define BLYNK_PRINT
Serial #include <WiFi.h>
#include <BlynkSimpleEsp32.h>
char auth[] =
"YOUR_BLYNK_TOKEN"; char ssid[]
= "YOUR_WIFI";
char pass[] = "PASSWORD";
int flowPin = 34;
int relayPin = 26;
void setup() {
Serial.begin(9600);
pinMode(relayPin,
OUTPUT);
digitalWrite(relayPin, LOW);
Blynk.begin(auth, ssid, pass);
}
void loop() {
Blynk.run();
int flowValue = analogRead(flowPin);
Serial.print("Flow: ");
Serial.println(flowValue);
// Leakage detected
if(flowValue > 2500) {
```

```

Serial.println("LeakageDetected!");
digitalWrite(relayPin, HIGH); // CLOSE valve
Blynk.logEvent("leak_alert", "Leakage detected!
Valve closed automatically"); delay(5000);
}
else {
digitalWrite(relayPin, LOW); // OPEN valve
}

```

ARDUINO CODE:

```

volatile int flow_frequency;
float flowRate;
unsigned char flowsensor =
2; int relayPin = 7; // valve
control
unsigned long currentTime;
unsigned long cloopTime;
void flow() {
flow_frequency++;
}
void setup() {
pinMode(flowsensor,
INPUT); pinMode(relayPin,
OUTPUT);
digitalWrite(relayPin, LOW); // valve open initially
Serial.begin(9600);
attachInterrupt(0, flow, RISING);

```

```

}
void loop() {
  currentTime = millis();
  if(currentTime >= (cloopTime + 1000)) {
    cloopTime = currentTime;
    flowRate = (flow_frequency / 7.5);
    Serial.print("Flow Rate: ");
    Serial.println(flowRate);
    // □ LEAKAGE DETECTED
    if(flowRate > 10) {
      Serial.println("Leakage Detected!");
      digitalWrite(relayPin, HIGH); // CLOSE valve
    }
    else {
      digitalWrite(relayPin, LOW); // OPEN valve
    }
    flow_frequency = 0;
  }
}

```

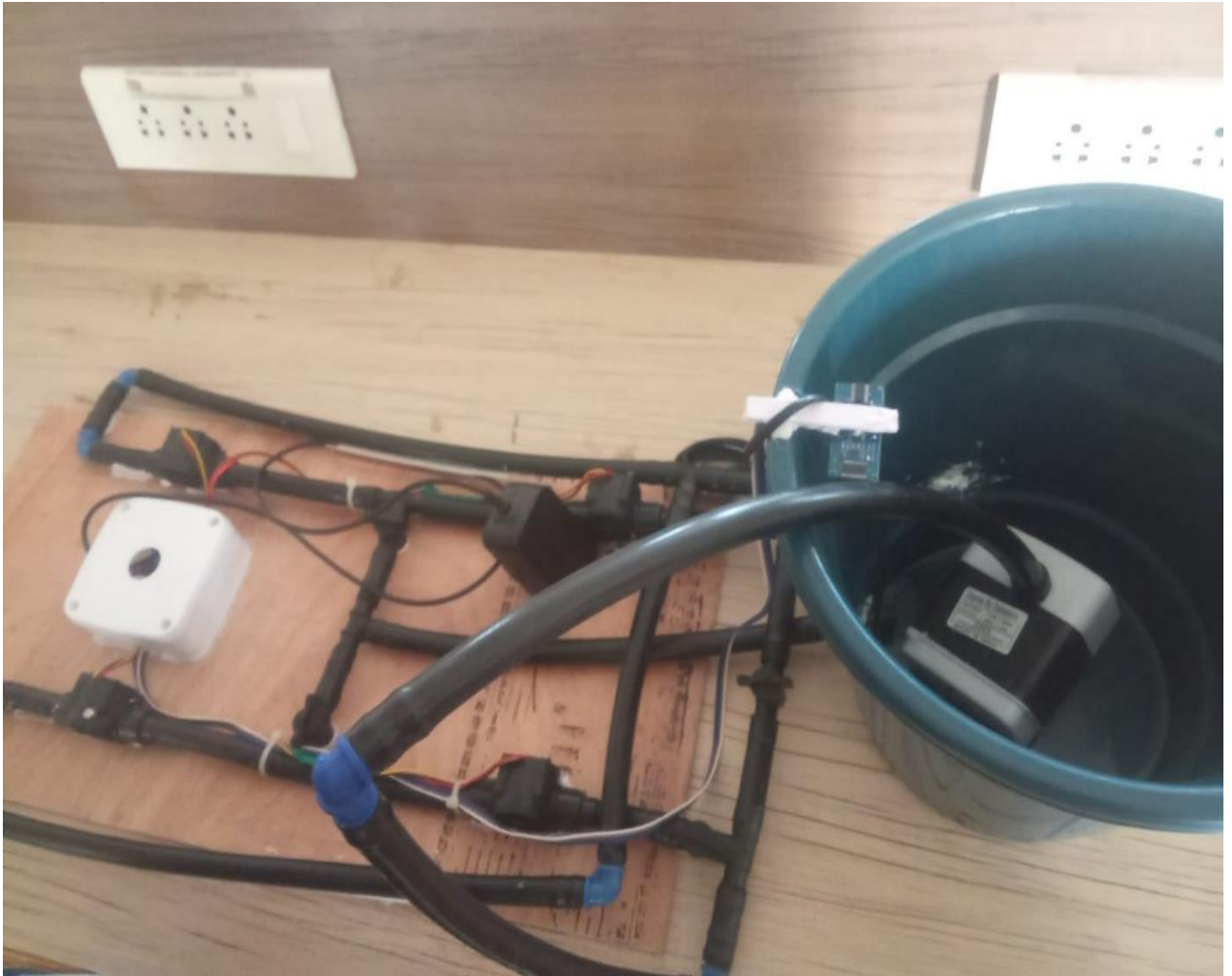
REFERENCES

- [1] M. Al-Zubi, K. Al-Fuqaha, and A. Al-Ali, “IoT-Based Smart Pipeline Monitoring System Using LoRaWAN Communication,” *IEEE Internet of Things Journal*, vol. 11, no. 2, pp. 1456–1465, 2024.
- [2] Y. Chen, H. Liu, and J. Wang, “Optical Fiber Speckle Pattern Leakage Detection System for Water Pipelines,” *IEEE Photonics Technology Letters*, vol. 35, no. 4, pp. 225–228, 2023.
- [3] S. Fulawade, P. Patil, and R. Kulkarni, “IoT-Based Smart Water Leakage Detection System Using Arduino UNO,” *IEEE Access*, vol. 11, pp. 45231–45240, 2023.
- [4] R. Gupta and P. Sharma, “Smart Water Distribution System Using IoT and Cloud Computing,” *IEEE Access*, vol. 11, pp. 29841–29850, 2023.
- [5] M. Johnson, T. Clarke, and D. Roberts, “Distributed Acoustic Sensing for Smart Pipeline Monitoring,” *IEEE Transactions on Instrumentation and Measurement*, vol. 73, pp. 1–10, 2024.
- [6] A. Kumar, S. Verma, and N. Singh, “IoT-Based Smart Water Leakage Control System with Automated Valve Operation,” *IEEE Sensors Letters*, vol. 8, no. 2, pp. 1–4, 2024.
- [7] Y. Li, X. Zhao, and F. Wang, “Low-Cost Fiber Optic Sensor for Water Pipeline Leakage Detection,” *IEEE Sensors Journal*, vol. 24, no. 3, pp. 2750–2758, 2024.
- [8] S. K. Patel and M. Desai, “IoT-Based Pipeline Leakage Detection Using Pressure Sensors,” *IEEE Transactions on Industrial Informatics*, vol. 19, no. 6, pp. 6401–6409, 2023.

- [9] J. Smith, A. Brown, and L. Wilson, “AI-Based Water Pipeline Leakage Detection System Using Machine Learning,” *IEEE Internet of Things Journal*, vol. 10, no. 8, pp. 7125–7134, 2023.
- [10] L. Zhang, “Fiber Optic Distributed Sensing System for Water Pipeline Monitoring,” *IEEE Sensors Journal*, vol. 23, no. 5, pp. 4890–4898, 2023.
- [11] H. Ahmed and R. Khan, “Wireless Sensor Network Based Smart Water Leakage Detection System,” *IEEE Access*, vol. 12, pp. 11234–11245, 2024.
- [12] P. Brown and T. Lewis, “Machine Learning Techniques for Real-Time Pipeline Leak Detection,” *IEEE Transactions on Automation Science and Engineering*, vol. 21, no. 1, pp. 210–219, 2024.
- [13] D. Garcia, M. Lopez, and S. Torres, “Cloud-Based Smart Water Monitoring Using IoT Sensors,” *IEEE Internet of Things Journal*, vol. 11, no. 5, pp. 3567–3576, 2024.
- [14] K. Singh and V. Rao, “Pressure and Flow Sensor Based Leakage Identification in Water Pipelines,” *IEEE Sensors Letters*, vol. 9, no. 1, pp. 15–19, 2025.
- [15] T. Wilson and J. Martin, “Hybrid AI and IoT Framework for Urban Water Leakage Management,” *IEEE Access*, vol. 13, pp. 7854–7865, 2025.

OUTCOME:

- Hardware Design



Position: First Prize in Project Presentation

Title: Event-Based Smart Water Pipeline Monitoring and Automatic Valve Control System

Date: 10 April 2026

Place: Bannari Amman Institute of Technology